

Quantum Field Theory I Exercise 5

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June 16, 2026

1 The beta function of Yukawa theory

1.1 The momentum-space 3-point correlation function

The momentum space 3-point correlation function $G^{(3)} = \langle \bar{\psi}(p_1)\psi(p_2)\phi(p_3) \rangle$ has a single first order contributing diagrams, which is the treelike single vertex diagram (figure 1). There are technically an additional tadpole diagram, but as $\langle \phi \rangle = 0$, we can ignore it. This gives the value (using momentum space Feynman rules for the correlation function):

$$G^{(3)} = \langle \bar{\psi}(p_1)\psi(p_2)\phi(p_3) \rangle = -ig + \mathcal{O}(g^3) \quad (1)$$

The g^3 correction has four diagrams:

- The 3-point triangle loop diagram
- Three loops extensions to the first order

These diagrams are diaplyed in figure 2.

1.2 The Callan-Symanzik equation

Using the renormalization factors

$$\phi_0 = \sqrt{Z_\phi}\phi \quad \psi_0 = \sqrt{Z_\psi}\psi \quad (2)$$

We can write the correlation function:

$$\langle \bar{\psi}_0(p_1)\psi_0(p_2)\phi_0(p_3) \rangle = \sqrt{Z_\phi}Z_\psi G^{(3)}(p_1, p_2, p_3; g(M), M) \quad (3)$$

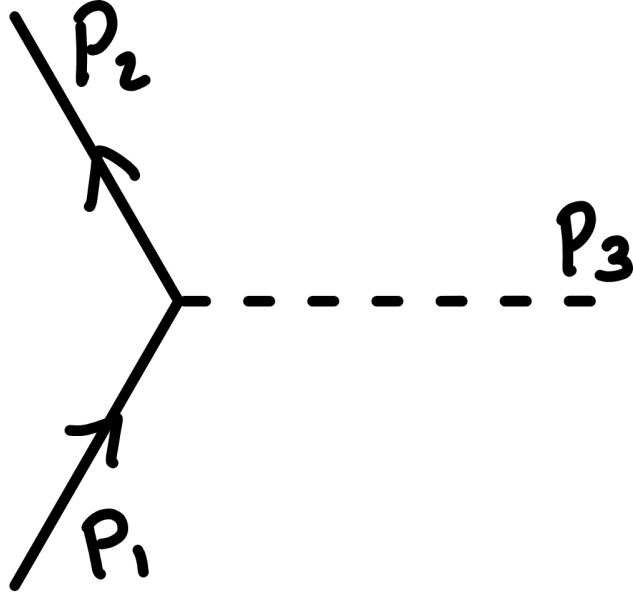


Figure 1: The treelike single vertex diagram.

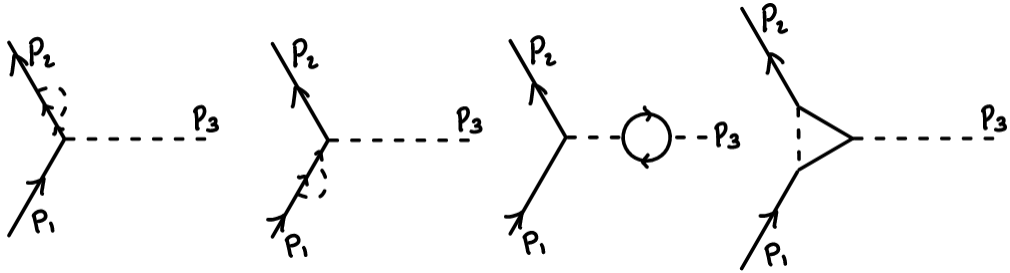


Figure 2: The four corrections of order g^3 . The rightmost one is the 3-point triangle loop diagram, while the other three are the loop extensions to the single vertex treelike diagram

Applying $M \frac{d}{dM}$ will give the Callan-Symanzik equation:

$$\left[\gamma_\phi + 2\gamma_\psi + \beta \partial_g + M \partial_M \right] G^{(3)} = 0 \quad (4)$$

Where

$$\gamma_\phi = \frac{1}{2Z_\phi} M \frac{dZ_\phi}{dM} \quad \gamma_\psi = \frac{1}{2} M \frac{dZ_\psi}{dM} \quad \beta = M \frac{dg}{dM} \quad (5)$$

1.3 Calculating γ_ϕ

For high energies $p^2 > m_\phi^2, m_\psi^2$, we can consider the theory as practically massless, and use exactly the same dimensional analysis we used in class:

$$G_\phi^{(2)}(p) = \frac{i}{p^2} - \frac{i}{p^2} \left(ip^2 \left(A \log \frac{\Lambda^2}{p^2} + C_1 \right) + iC_2 \Lambda^2 \right) \frac{i}{p^2} + \frac{i}{p^2} (ip^2 \delta_{z_\phi} - i\delta_{m_\phi}) \frac{i}{p^2} \quad (6)$$

Which means:

$$\Sigma(p^2) = p^2 \left(A \log \frac{\Lambda^2}{p^2} + C_1 \right) - C_2 \Lambda^2 - p^2 \delta_{z_\phi} + \delta_{m_\phi} \quad (7)$$

And the renormalization condition

$$\left. \frac{d}{dp^2} \Sigma \right|_{p^2=M^2} = 0 \quad (8)$$

Gives:

$$A \log \frac{\Lambda^2}{M^2} - A + C_1 - \delta_{z_\phi} = 0 \quad (9)$$

$$\delta_{z_\phi} = A \log \frac{\Lambda^2}{M^2} - A + C_1 \quad (10)$$

Therefore, to find γ_ϕ we can write:

$$\frac{d\delta_{z_\phi}}{dM} = A \frac{1}{\frac{\Lambda^2}{M^2}} \frac{-2\Lambda^2}{M^3} = \frac{-2A}{M} \quad (11)$$

$$\gamma_\phi = \frac{1}{2} M \frac{d\delta_{z_\phi}}{dM} = -A \quad (12)$$

As we saw in the tutorial about Yukawa propegators:

$$\Sigma(p^2) \supset \frac{3g^2}{4\pi^2} \int_0^1 dx \left[(m_\psi^2 - x(1-x)p^2) \log \left(\frac{1}{m_\psi^2 - x(1-x)p^2} \right) \right] \quad (13)$$

Which in the high energy limit $m_\psi \ll p$ gives:

$$\Sigma(p^2) \supset -\frac{3g^2}{4\pi^2} \int_0^1 dx x(1-x)p^2 \log \left(\frac{1}{p^2} \right) = -\frac{g^2}{8\pi^2} p^2 \log \left(\frac{1}{p^2} \right) \quad (14)$$

And so we get:

$$\gamma_\phi = \frac{g^2}{8\pi^2} \quad (15)$$

1.4 Calculating γ_ψ

We again consider the theory as massless, and write the Fermion propegator as we saw in the same tutorial:

$$G_\psi^{(2)}(p) = \frac{i}{\not{p} - \tilde{\Sigma}(p) + i\epsilon} \quad (16)$$

Where

$$\begin{aligned} \tilde{\Sigma}(p) \supset & -\frac{g^2}{16\pi^2} \int_0^1 dx \left[(m_\psi + x\not{p}) \log \frac{1}{-x(1-x)p^2 + xm_\phi^2 + (1-x)m_\psi^2} \right] \\ & \xrightarrow{m_\phi, m_\psi \ll M} -\frac{g^2}{16\pi^2} \int_0^1 x dx \not{p} \log \frac{1}{p^2} = -\frac{g^2}{32\pi^2} \not{p} \log \frac{1}{p^2} \end{aligned} \quad (17)$$

Using again the same logic as in the scalar case, in the massless case we can use dimensional arguments to get a similar propegator up to second order in g :

$$G_\psi^{(2)}(p) = \frac{i}{\not{p}} - \frac{i}{\not{p}} \left(i\not{p} \left(A \log \frac{\Lambda^2}{p^2} + C_1 \right) + iC_2 \Lambda \right) \frac{i}{\not{p}} + \frac{i}{\not{p}} \left(i\not{p} \delta_{z_\psi} - i\delta_{m_\psi} \right) \frac{i}{\not{p}} \quad (18)$$

Giving:

$$\tilde{\Sigma}(p) = \not{p} \left(A \log \frac{\Lambda^2}{p^2} + C_1 \right) - C_2 \Lambda - \not{p} \delta_{z_\psi} + \delta_{m_\psi} \quad (19)$$

We can write it as:

$$\tilde{\Sigma}(p) = \not{p} \tilde{\Sigma}^{(0)}(p^2) + \tilde{\Sigma}^{(1)}(p^2) \quad (20)$$

Where

$$\tilde{\Sigma}^{(0)}(p^2) = \left(A \log \frac{\Lambda^2}{p^2} + C_1 \right) - \delta_{z_\psi} \quad \tilde{\Sigma}^{(1)}(p^2) = -C_2 \Lambda + \delta_{m_\psi} \quad (21)$$

Using the second renormalization condition:

$$\tilde{\Sigma}^{(0)}(M^2) = 0 \quad (22)$$

Gives:

$$\delta_{z_\psi} = A \log \frac{\Lambda^2}{M^2} + C_1 \quad (23)$$

This will again give:

$$\gamma_\psi = \frac{1}{2} M d \frac{\delta_{z_\psi}}{dM} = -A = \frac{g^2}{32\pi^2} \quad (24)$$

1.5 Finiding the β function

The 1-loop correction to the 3 point function is a result of the four diagrams we saw above. Each diagram has 3 vertices, three internal propegators (1 scalar and 2 fermions), and one loop giving undetermined momentum.

Three of the diagrams are just 1-loop correations to the signle vertex diagram. Using the two 1-loop corrections we found in the last two sections, and multiplying them by the additional $(-ig)$ factor from the vertex, gives

$$\begin{aligned} G_1^{(3)} &\supset i \frac{g^3}{32\pi^2} \bar{v}(p_2) u(p_1) \log \frac{\Lambda^2}{M^2} \\ G_2^{(3)} &\supset i \frac{g^3}{32\pi^2} \bar{v}(p_2) u(p_1) \log \frac{\Lambda^2}{M^2} \\ G_3^{(3)} &\supset i \frac{g^3}{8\pi^2} \bar{v}(p_2) u(p_1) \log \frac{\Lambda^2}{M^2} \end{aligned} \quad (25)$$

The last diagram is the triangle loop diagram:

$$(-ig)^3 \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m_\phi^2 + i\epsilon} \bar{v}(p_2) \left[\frac{i((p-p_1)_\mu \gamma^\mu + m_\psi)}{(p-p_1)^2 - m_\psi^2 + i\epsilon} \frac{i((p+p_3)_\mu \gamma^\mu - m_\psi)}{(p+p_3)^2 - m_\psi^2 + i\epsilon} \right] u(p_1) \quad (26)$$

This integral goes like $\int \frac{p^2 d^4 p}{p^6}$ which diverges logarithmically. So, let us use dimensional regularization.

$$\begin{aligned}
& (-ig)^3 \int \frac{d^d p}{(2\pi)^d} \frac{i}{p^2 - m_\phi^2 + i\epsilon} \bar{v}(p_2) \left[\frac{i((p-p_1)_\mu \gamma^\mu + m_\psi)}{(p-p_1)^2 - m_\psi^2 + i\epsilon} \frac{i((p+p_3)_\mu \gamma^\mu - m_\psi)}{(p+p_3)^2 - m_\psi^2 + i\epsilon} \right] u(p_1) \supset \\
& \supset g^3 \int \bar{v}(p_2) u(p_1) \frac{d^d p}{(2\pi)^d} \frac{p^2}{[p^2 - m_\phi^2 + i\epsilon] [(p-p_1)^2 - m_\psi^2 + i\epsilon] [(p+p_3)^2 - m_\psi^2 + i\epsilon]}
\end{aligned} \tag{27}$$

Where we took only the logarithmically divergent part of the nominator, and pulled the spinors outside the integral. Now, using Feynman's trick with two variables:

$$\begin{aligned}
& \int \frac{d^d p}{(2\pi)^d} \frac{p^2}{[p^2 - m_\phi^2 + i\epsilon] [(p-p_1)^2 - m_\psi^2 + i\epsilon] [(p+p_3)^2 - m_\psi^2 + i\epsilon]} = \\
& = \int \frac{d^d p}{(2\pi)^d} \int_0^1 dx dy dz \delta(x+y+z-1) \times \\
& \times \frac{p^2}{([p^2 - m_\phi^2 + i\epsilon] x + [(p-p_1)^2 - m_\psi^2 + i\epsilon] y + [(p+p_3)^2 - m_\psi^2 + i\epsilon] z)^3} = \\
& = \int \frac{d^d p}{(2\pi)^d} \int_0^1 dx dy dz \delta(x+y+z-1) \times \\
& \times \frac{p^2}{((x+y+z)p^2 + 2(p_3 z - p_1 y)p + (p_3^2 z + p_1^2 y) - m_\phi^2 x - m_\psi^2(y+z))^3} = \\
& = \int \frac{d^d p}{(2\pi)^d} \int_0^1 dy \int_0^{1-y} dz \times \\
& \times \frac{p^2}{(p^2 + 2(p_3 z - p_1 y)p + (p_3^2 z + p_1^2 y) - m_\phi^2 + (m_\psi^2 - m_\psi^2)(y+z))^3}
\end{aligned} \tag{28}$$

It is simple now to complete the square:

$$l = p + p_3 z - p_1 y \quad \Delta = (p_3 z - p_1 y)^2 - (p_3^2 z + p_1^2 y) - m_\phi^2 + (m_\psi^2 - m_\psi^2)(y+z) \tag{29}$$

To give:

$$\begin{aligned}
& \int \frac{d^d p}{(2\pi)^d} \int_0^1 dy \int_0^{1-y} dz \times \\
& \times \frac{p^2}{\left(p^2 + 2(p_3 z - p_1 y)p + (p_3^2 z + p_1^2 y) - m_\phi^2 + (m_\psi^2 - m_\psi^2)(y + z)\right)^3} \supset \\
& \supset \int_0^1 dy \int_0^{1-y} dz \int \frac{d^d l}{(2\pi)^d} \frac{l^2}{(l^2 - \Delta)^3}
\end{aligned} \tag{30}$$

Where in the last step we again took only at the part of the nominator that gives l^2 , to get the logarithmic divergence. Using the known integral (in Euclidean space):

$$\begin{aligned}
& \int_0^1 dy \int_0^{1-y} dz \int \frac{d^d l}{(2\pi)^d} \frac{l^2}{(l^2 - \Delta)^3} = \\
& = i \int_0^1 dy \int_0^{1-y} dz \int \frac{d^d l_E}{(2\pi)^d} \frac{l^2}{(l_E^2 + \Delta)^3} = \\
& = i \int_0^1 dy \int_0^{1-y} dz \frac{d \Gamma(2 - d/2)}{2} \frac{1}{2(4\pi)^{d/2}} \Delta^{-2+d/2}
\end{aligned} \tag{31}$$

Taking $d = 4 - \epsilon$:

$$\begin{aligned}
& i \frac{g^3}{16\pi^2} \bar{v}(p_2) u(p_1) \Gamma(\epsilon/2) \int_0^1 dy \int_0^{1-y} dz \Delta^{-\epsilon/2} = \\
& = i \frac{g^3}{16\pi^2} \bar{v}(p_2) u(p_1) \int_0^1 dy \int_0^{1-y} dz \times \\
& \times \left(\frac{2}{\epsilon} - \gamma - \log \frac{1}{(p_3 z - p_1 y)^2 - (p_3^2 z + p_1^2 y) - m_\psi^2 + (m_\psi^2 - m_\psi^2)(y + z)} \right)
\end{aligned} \tag{32}$$

Now, selecting $p_3^2 = M^2$ and $p_1^2 = p_2^2 = m_\psi^2 \ll M^2$ we can write the logarithmically divergent part as:

$$i \frac{g^3}{16\pi^2} \bar{v}(p_2) u(p_1) \int_0^1 dy \int_0^{1-y} dz \log \frac{1}{M^2} = i \frac{g^3}{32\pi^2} \bar{v}(p_2) u(p_1) \log \frac{1}{M^2} \tag{33}$$

So, the total logarithmic divergence can be written, for some cutoff Λ , as:

$$G^{(3)}(p_1, p_2, p_3) \supset \left(\frac{i^3}{p_1^2 p_2^2 p_3^2} \right) i \frac{7}{32\pi^2} g^3 \bar{v}(p_2) u(p_1) \log \frac{\Lambda^2}{M^2} (2\pi)^4 \delta^4(p_3 - p_1 - p_2) \tag{34}$$

Adding the first order and the counter term $g_0 = g + \delta_g$:

$$G^{(3)}(p_1, p_2, p_3) \supset \left(\frac{i^3}{p_1^2 p_2^2 p_3^2} \right) \bar{v}(p_2) u(p_1) \left(ig + i\delta_g + i \frac{7}{32\pi^2} g^3 \log \frac{\Lambda^2}{M^2} \right) (2\pi)^4 \delta^4(p_3 - p_1 - p_2) \quad (35)$$

Requirng the renormalization condition:

$$\Sigma(p_1^2 = p_2^2 = m_\phi^2, p_3^2 = M^2) = 0 \quad (36)$$

Gives:

$$\delta_g = - \frac{7}{32\pi^2} g^3 \log \frac{\Lambda^2}{M^2} \quad (37)$$

Which finally gives:

$$\beta = M \partial_M \delta_g = \frac{7}{16\pi^2} g^3 \quad (38)$$

2 The beta function at small M^2

Let us define:

$$\lambda_p(M) = -\mathcal{M} \quad s = t = u = M^2 \quad (39)$$

This is enough for us to note that in ϕ^4 the $2 \rightarrow 2$ scattering amplitude for a generic momentum (up to second order in λ_p):

$$\begin{aligned} -\mathcal{M}(p) &= \lambda_p + \lambda_p^2 (V(s) + V(t) + V(u) - 3V(M^2)) = \\ &= \lambda_p + \frac{\lambda_p^2}{32\pi^2} \int_0^1 dx \sum_{p^2=s,t,u} \ln \left(\frac{m^2 - x(1-x)p^2}{m^2 - x(1-x)M^2} \right) \end{aligned} \quad (40)$$

So, the β function is:

$$\begin{aligned} \beta &= M \partial_M \lambda = M \partial_M (-\mathcal{M}) = \\ &= \frac{iM\lambda_p^2}{32\pi^2} \partial_M \int_0^1 dx \sum_{p^2=s,t,u} \ln \left(\frac{m^2 - x(1-x)p^2}{m^2 - x(1-x)M^2} \right) = \\ &= -\frac{3M\lambda_p^2}{32\pi^2} \int_0^1 dx \partial_M \ln (m^2 - x(1-x)M^2) = \\ &= -\frac{3M\lambda_p^2}{32\pi^2} \int_0^1 dx \frac{2x(1-x)M}{m^2 - x(1-x)M^2} \end{aligned} \quad (41)$$

Taking the $M^2 \ll m^2$ limit gives:

$$\beta = -\frac{3\lambda_p^2}{16\pi^2} \int_0^1 dx x(1-x) \frac{M^2}{m^2} \quad (42)$$

$$\beta = -\frac{\lambda_p^2}{32\pi^2} \frac{M^2}{m^2} \quad (43)$$